



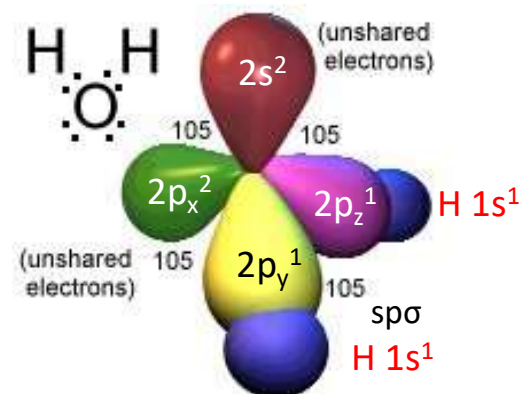
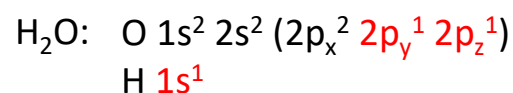
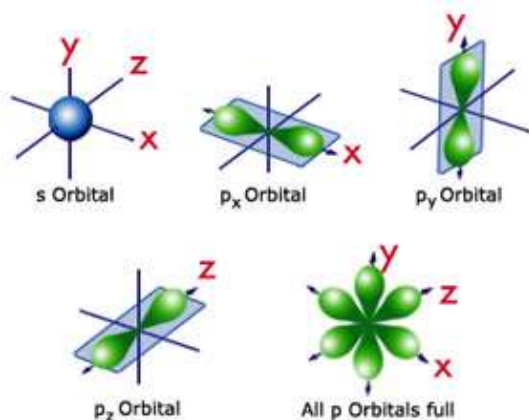
GEST 3015 Smart Materials: Structures and Properties

L2 HYBRIDIZATION

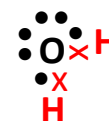
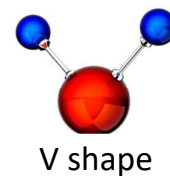
DR. MARTIN MA

Complex molecules - Valence bond (VB) theory

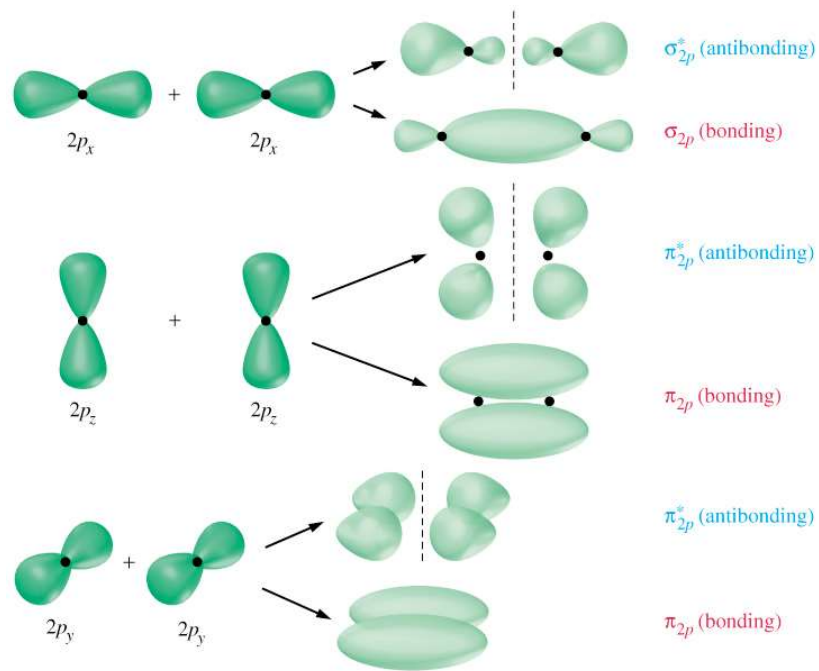
- The geometry of complex molecule depends on the wave functions of the outer electrons of their atoms
- Valence shell: outermost shell
- Only **valence electrons** in the valence shell involve inter-atom bonding in a molecule
- To determine molecular structures



Tetrahedral electron-orbital geometry



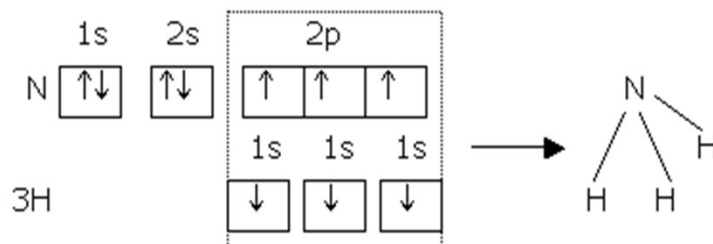
VB theory



- Location of the overlap region with respect to the bond axis
- **σ bond** overlaps along the bond axis (head-to-head)
- **π bond** overlaps in parallel to the bond axis (shoulder-to-shoulder)

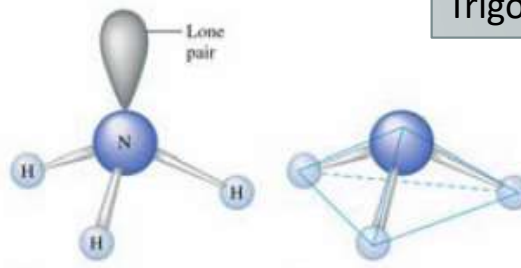
Exercise

What is the electron-pair geometry and molecule geometry of NH_3 ?

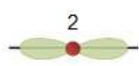
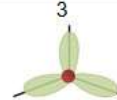


Electron-pair geometry:
Tetrahedral

Molecular geometry:
Trigonal pyramidal



Electron-pair geometry and molecule geometry

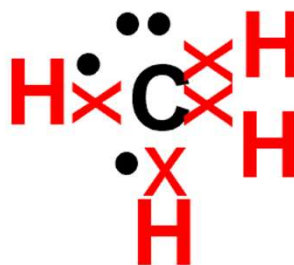
Number of Electron Dense Areas	Electron-Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2 	Linear	 Linear				
3 	Trigonal planar	 Trigonal planar	 Bent			
4 	Tetrahedral	 Tetrahedral	 Trigonal pyramidal	 Bent		
5 	Trigonal bipyramidal	 Trigonal bipyramidal	 Sawhorse	 T-shaped	 Linear	
6 	Octahedral	 Octahedral	 Square pyramidal	 Square planar	 T-shaped	 Linear

Hybrid orbitals – sp^3 hybridization

How about CH_4 (Methane)?

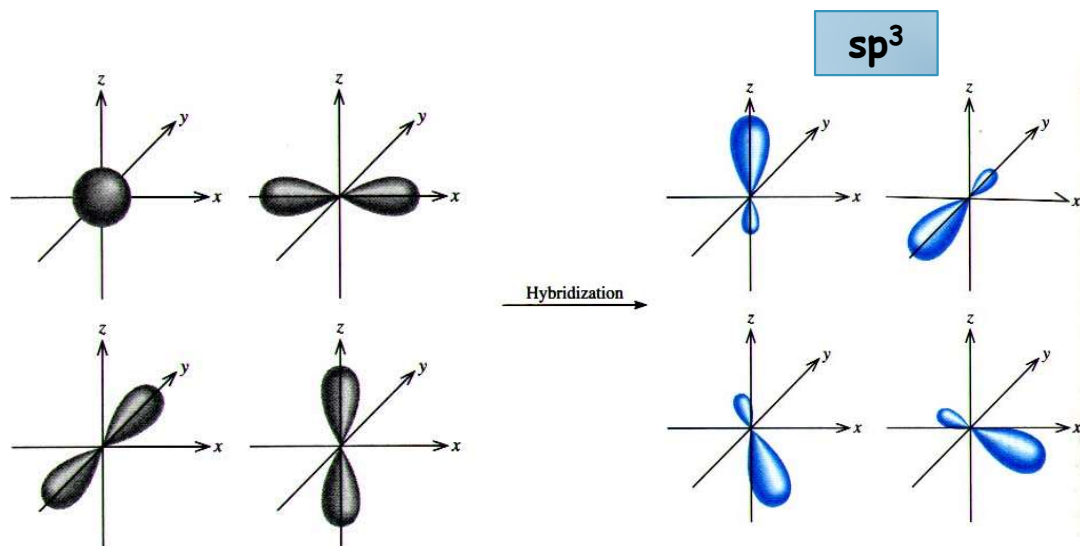
C: $1s^2 2s^2 (2p_x^1 2p_y^1 2p_z^0)$

H: $1s^1$



Hybrid orbitals – sp^3 hybridization

Hybrid orbitals consist of mixtures of s and p orbitals.



The wave functions of these hybrid orbitals are:

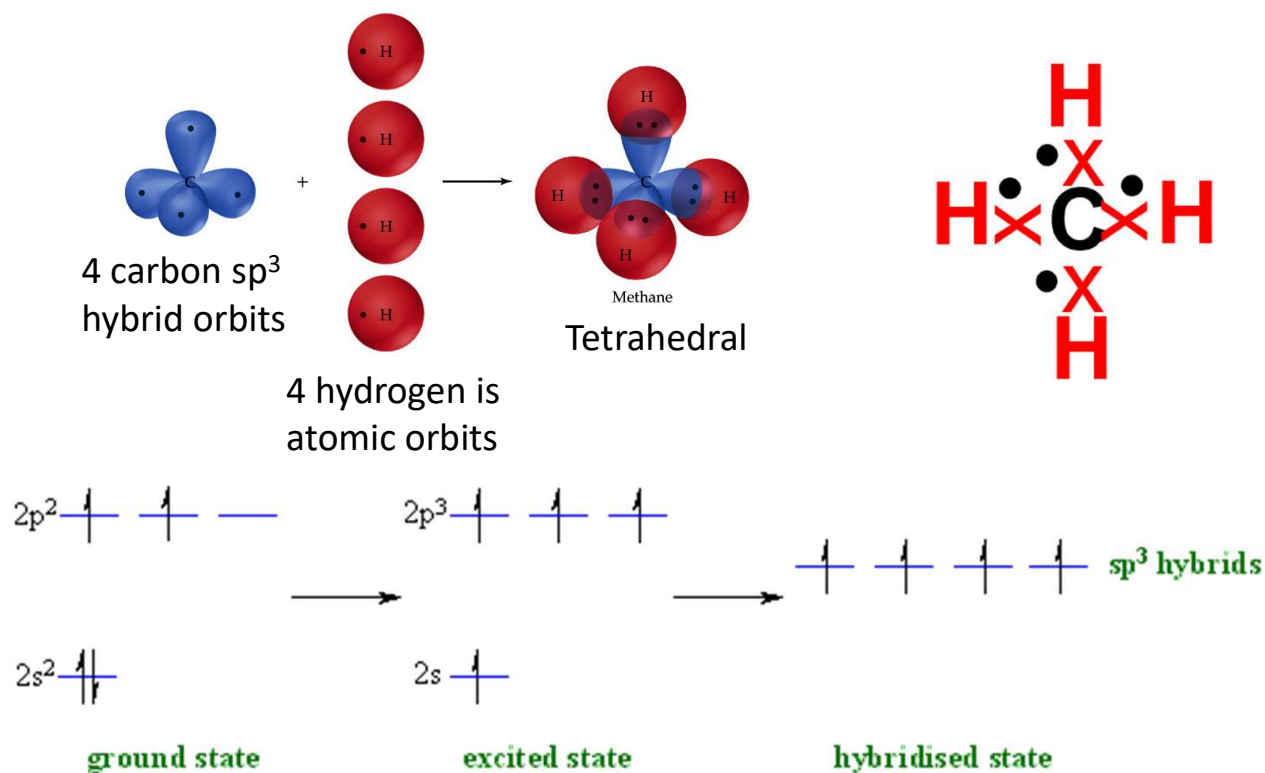
$$\psi_1 = \frac{1}{2}[\psi_s + \psi_{p_x} + \psi_{p_y} + \psi_{p_z}]$$

$$\psi_2 = \frac{1}{2}[\psi_s + \psi_{p_x} - \psi_{p_y} - \psi_{p_z}]$$

$$\psi_3 = \frac{1}{2}[\psi_s - \psi_{p_x} - \psi_{p_y} + \psi_{p_z}]$$

$$\psi_4 = \frac{1}{2}[\psi_s - \psi_{p_x} + \psi_{p_y} - \psi_{p_z}]$$

Hybrid orbitals – sp^3 hybridization

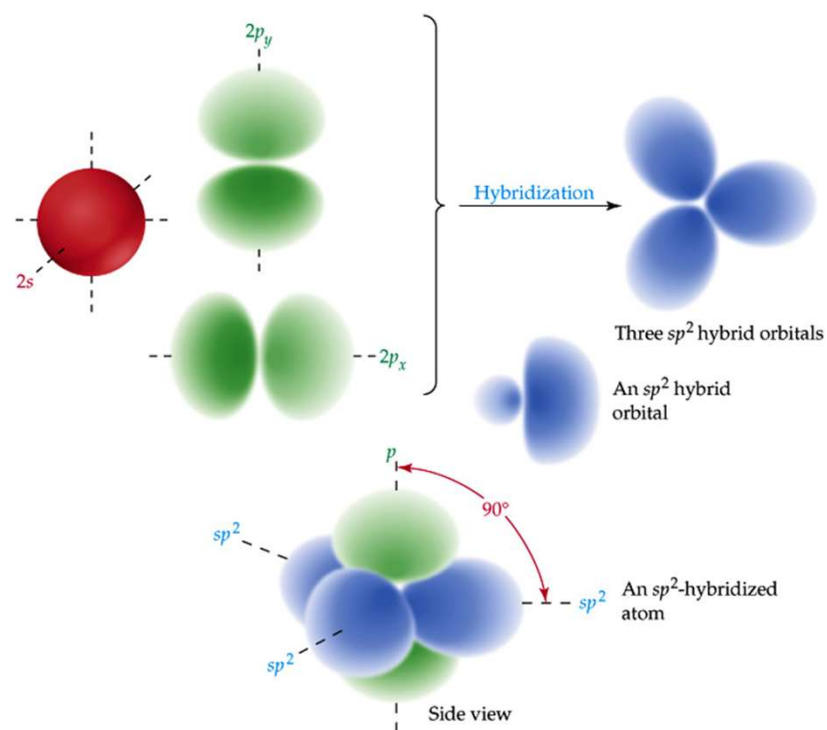


Hybrid orbitals - sp^2 hybridization

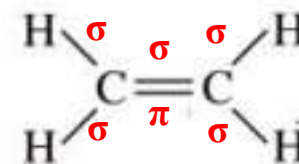
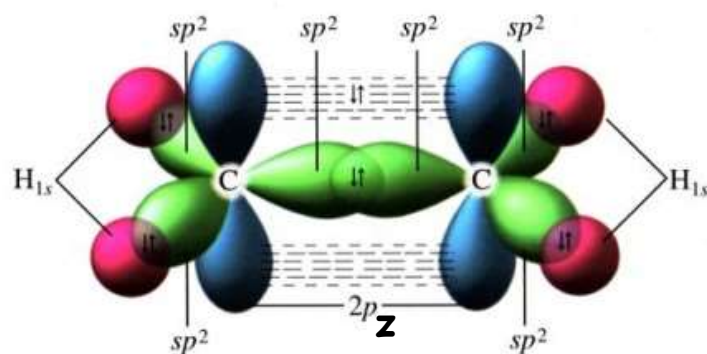
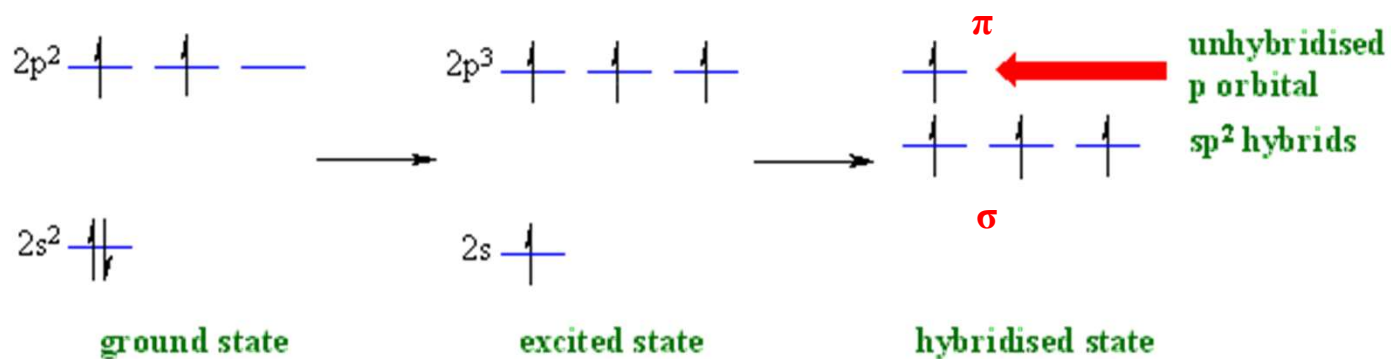
How about C_2H_4 (Ethene)?

C: $1s^2 2s^2 (2p_x^1 2p_y^1 2p_z^0)$

H: $1s^1$



Hybrid orbitals - sp^2 hybridization

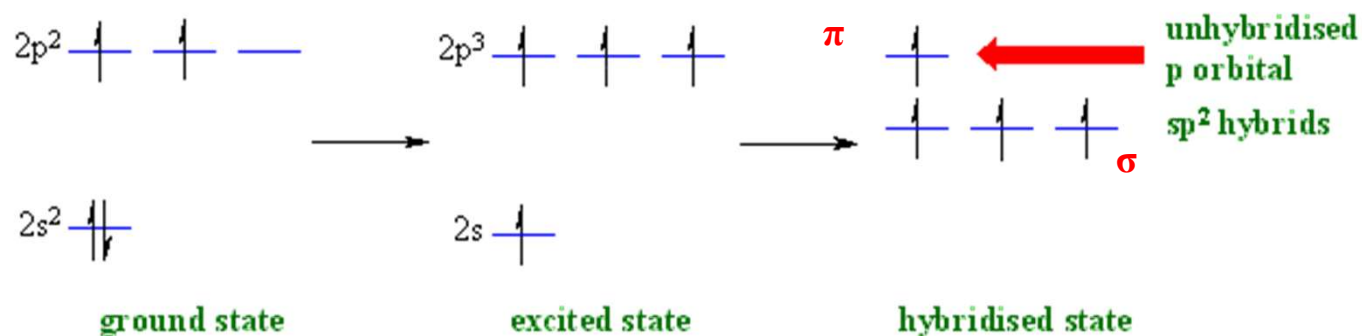
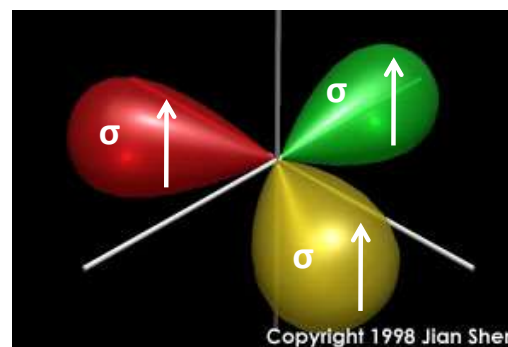


Hybrid orbitals – sp^2 hybridization

How about C_6H_6 (Benzene) ?

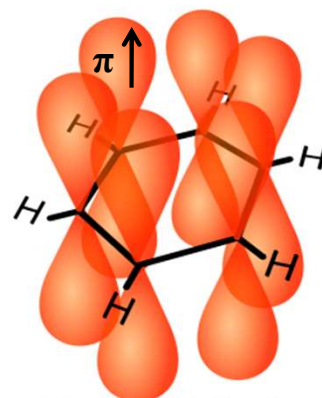
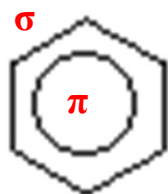
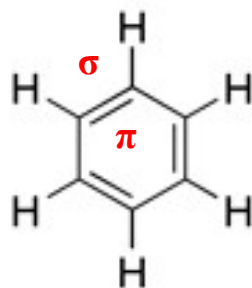
C: $1s^2 2s^2 (2p_x^1 2p_y^1 2p_z^0)$

H: $1s^1$

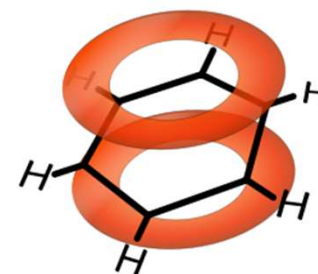


Hybrid orbitals – sp^2 hybridization

C_6H_6 (Benzene)



6 p-orbitals



delocalized

The total of six 2p orbitals in the molecule combine into bonding π orbitals that are continuous above and below the plane of the ring.

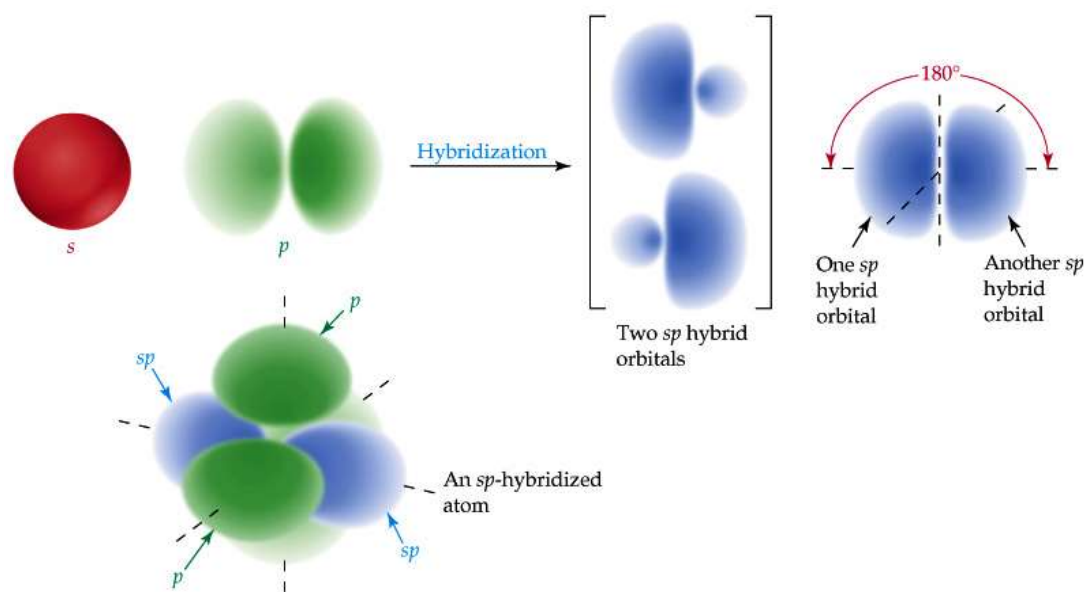
The six electrons involved belong to the molecule as a whole and not to any particular pair of atoms; these electrons are delocalized.

Hybrid orbitals –sp hybridization

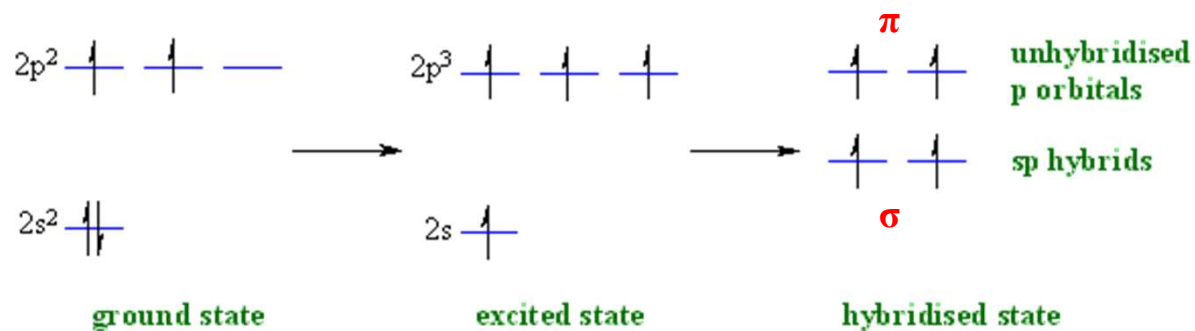
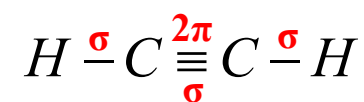
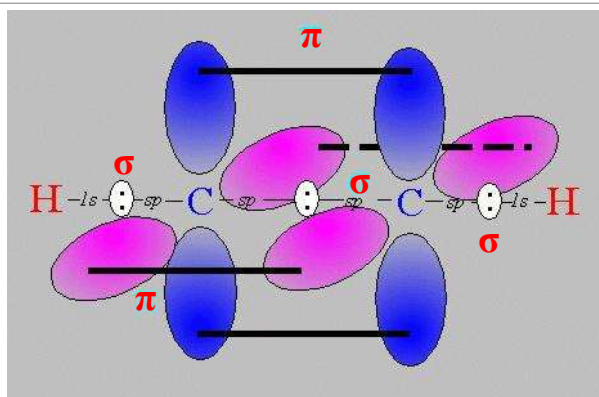
How about C₂H₂(Ethyne) ?

C: 1s² 2s² (**2p_x¹ 2p_y¹ 2p_z⁰**)

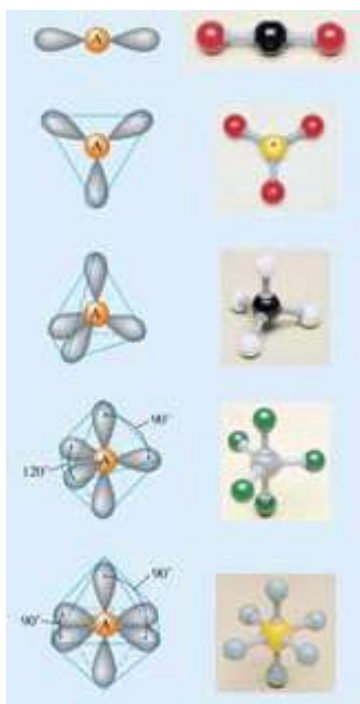
H: **1s¹**



Hybrid orbitals –sp hybridization



Orbital hybridization



sp (linear)

sp^2 (trigonal planar)

sp^3 (tetrahedral)

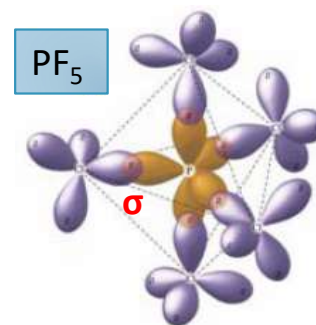
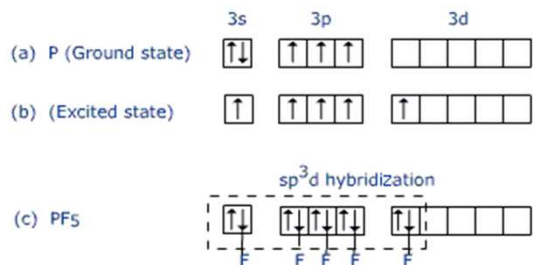
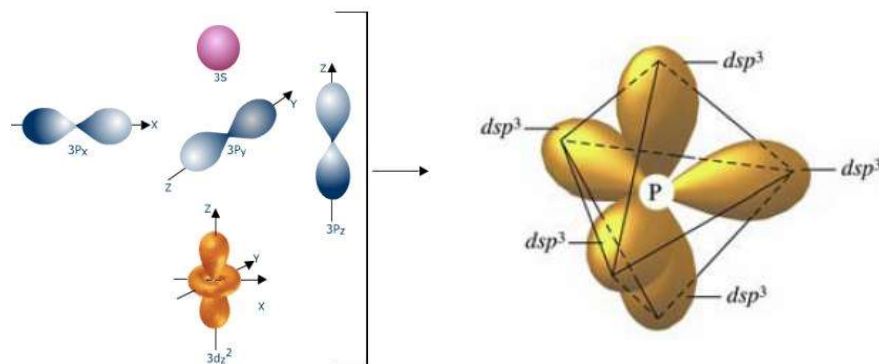
sp^3d

sp^3d^2

Hybrid orbitals - PF₅

P $1s^2 2s^2 2p^6$ **$3s^2 3p_x^1 3p_y^1 3p_z^1 3d_{z^2}^0$** $3d_{x^2-y^2}^0 3d_{xy}^0 3d_{xz}^0 3d_{yz}^0$
 F $1s^2 2s^2 2p_x^2 2p_y^2$ **$2p_z^1$**

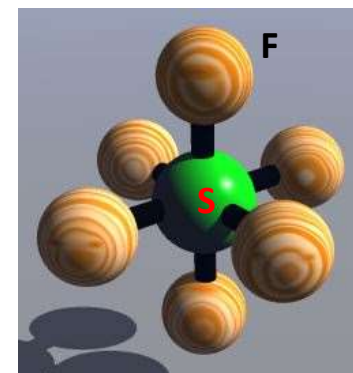
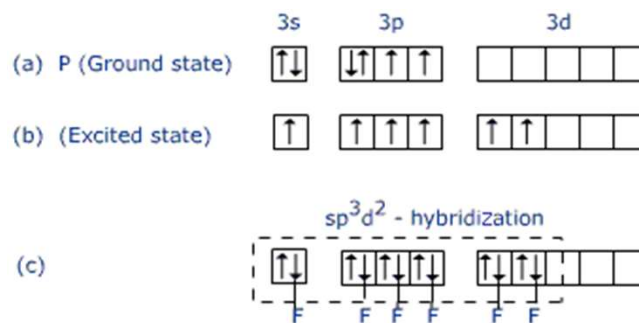
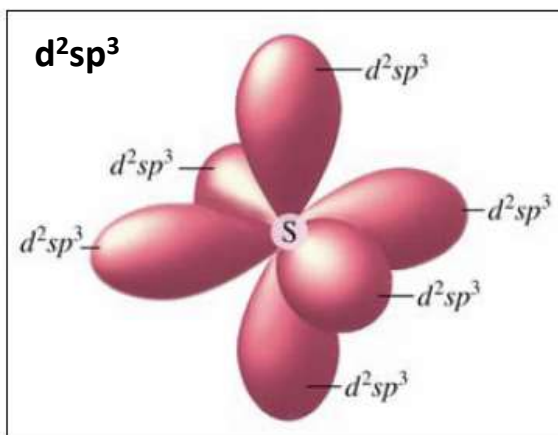
dsp^3 or sp^3d



Trigonal bipyramidal

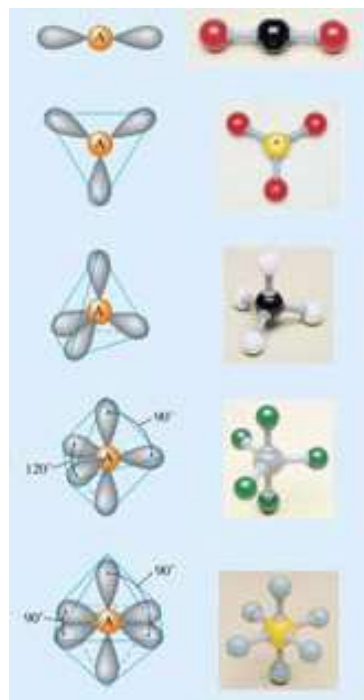
Hybrid orbitals - SF₆

S $1s^2 2s^2 2p^6$ **$3s^2 3p_x^2 3p_y^1 3p_z^1 3d_{x^2-y^2}^0 3d_{z^2}^0$** $3d_{xy}^0 3d_{xz}^0 3d_{yz}^0$
 F $1s^2 2s^2 2p_x^2 2p_y^2$ **$2p_z^1$**



Octahedral

Orbital hybridization



sp (linear)

sp^2 (trigonal planar)

sp^3 (tetrahedral)

sp^3d (trigonal bipyramidal)

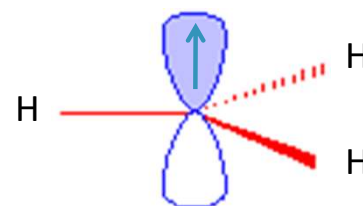
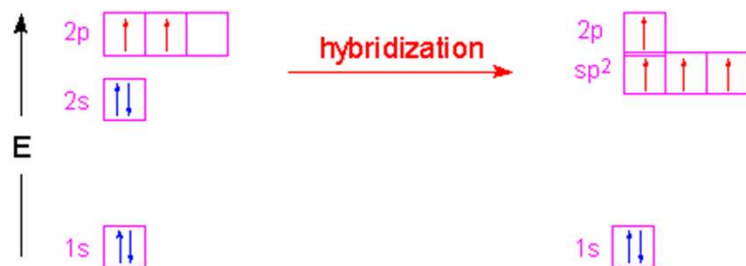
sp^3d^2 (octahedral)

Exercise

What is the molecular geometry of XeF_4 ? Does it have a dipole moment?

Excise

What is the molecular geometry of $\cdot\text{CH}_3$?



Trigonal planar

~END~